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Citation: Applied Physics Letters **107**, 043905 (2015); doi: 10.1063/1.4927741
View online: http://dx.doi.org/10.1063/1.4927741
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Optical properties of amorphous and polycrystalline Sb_2Se_3 thin films prepared by thermal evaporation

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(Received 11 May 2015; accepted 14 July 2015; published online 30 July 2015)

Antimony triselenide

Sb_2Se_3 is a very promising photovoltaic material because of its attractive material, optical and electrical properties. Very recently, we reported a superstrate CdS/Sb_2Se_3 solar cell with 5.6% certified efficiency. In this letter, we focused on the optical properties of amorphous and polycrystalline Sb_2Se_3 thin films prepared by thermal evaporation. Using temperature dependent transmission spectrum and temperature dependent photoluminescence, the indirect optical transition nature and bandgap values as functions of temperature were acquired. Using ellipsometry measurements and Swanepoel's envelope method, the refractive indices as well as the dielectric constant in a wide wavelength range of 193–2615 nm were obtained. These works would lay the foundation for the further development of Sb_2Se_3 thin film solar cells. © *2015 AIP Publishing LLC*.
[http://dx.doi.org/10.1063/1.4927741]

Antimony selenide (Sb_2Se_3) is a promising absorber material for photovoltaic application due to its properties such as (1) high absorption coefficient ($>10^5\,cm^{-1}$) at wavelength close to the absorption onset;[1] (2) approximately 1.1 eV (Ref. 2) bandgap optimal for single junction solar cells; (3) simple binary composition with fixed orthorhombic phase;[3] and (4) low toxicity and relatively earth-abundant constituents. All these features merit its exploration for photovoltaic application.[1,4–11] Recently, our group fabricated both superstrate[1,8,11–13] and substrate[7] thin film solar cells using Sb_2Se_3 as the absorption layer, and achieved 5.6% certified efficiency. To better design the device configuration and further improve device performance, a systematic understanding of the optical properties of Sb_2Se_3 is crucial. In addition, amorphous Sb_2Se_3 (a-Sb_2Se_3) has a larger bandgap than polycrystalline Sb_2Se_3 (c-Sb_2Se_3)[14–16] and could be potentially applied as the back surface field for further efficiency improvement, in analogy to the integration of amorphous Si with crystalline Si to build high efficiency heterojunction with intrinsic thin-layer solar cells.[17] To this end, the optical properties of both a-Sb_2Se_3 and c-Sb_2Se_3 thin films are worth investigating.

However, the optical transition type of Sb_2Se_3 is still controversial until so far. For Sb_2Se_3 single crystals and polycrystalline thin films, both direct and indirect transition type with bandgap (E_g) ranging from 1.0 to 1.2 eV were reported.[2,9,18] For a-Sb_2Se_3, optical edge (E_0) ranging from 1.2 to 1.5 eV was reported.[15,18] Other optical properties of Sb_2Se_3 thin films, such as the complex refractive index and the complex dielectric constant, are also fundamental physical constants and deserve to be thoroughly investigated. In the early publications, the refractive indices of a-Sb_2Se_3 and c-Sb_2Se_3 were investigated utilizing Swanepoel's envelope method[21] in the wavelength range of 800–2500 nm.[14,19,20] But the optical constants of Sb_2Se_3 thin films in the wavelength range of 200–800 nm are more important where most absorption occurs for photovoltaics. The relative dielectric constant of c-Sb_2Se_3, 13.7–26 at the infinite frequency and 80–133 at zero frequency, has been reported by different groups.[22–25] To conclude, the optical properties in the wavelength range of 200–2500 nm need to be carefully studied.

In this letter, we confirmed the optical transition type of c-Sb_2Se_3 by the temperature dependent transmission spectrum (TDTS) and temperature dependent photoluminescence (TDPL). We found that c-Sb_2Se_3 is an intrinsically indirect bandgap ($E_{g,\,i}$) material with the lowest $E_{g,i}$ of 1.03 ± 0.01 eV and the lowest direct bandgap ($E_{g,\,d}$) of 1.17 ± 0.02 eV at 300 K. The basic optical constants of a-Sb_2Se_3 and c-Sb_2Se_3 such as complex refractive index, complex dielectric constants, and optical reflectivity were characterized through a combined ellipsometry (EM) and transmission spectra measurements in the 193–2615 nm wavelength range, the spectrum most relevant for photovoltaic application. The measured large refractive and dielectric values suggest that the exciton binding energy in Sb_2Se_3 might be relatively small and an antireflective coating is recommended to enhance the light absorption of Sb_2Se_3 thin film solar cells.

A-Sb_2Se_3 and c-Sb_2Se_3 thin films were prepared on quartz substrates utilizing thermal evaporation with the substrates kept at room temperature and 290 °C, respectively. The substrate temperature was the only variable during depositing the two kinds of films. The deposition details could be found elsewhere.[7,8] X-ray diffraction (XRD) carrying out on Philips diffractometer (X pert pro MRD instrument) was applied to characterize the structure of Sb_2Se_3

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thin films (Fig. 1(a)). For Sb_2Se_3 film deposited with the substrates at room temperature, no diffraction peaks were observed, indicating the as-prepared film was amorphous. However, for Sb_2Se_3 film deposited with the substrates at 290 °C, strong diffraction peaks were observed, and all these peaks agreed well in position with the standard diffraction pattern of orthorhombic Sb_2Se_3 (JCPDS 00–015-0861), confirming their polycrystalline nature. Since structural reorganization of Sb_2Se_3 film starts at 133 °C,[20] we believe that the substrate temperature is decisive for the amorphous to polycrystalline transition. Film morphology was characterized using scanning electron microscopy (SEM) (FEI Nova NanoSEM450 microscope) and the corresponding top-view images were shown as insets in Fig. 1(a). For a-Sb_2Se_3, the film was composed of numerous small clusters; for c-Sb_2Se_3, the film displayed obvious crystalline grains with the biggest size over 2 μm and sharp grain boundary. An estimate of the mean crystal grain size in c-Sb_2Se_3 from XRD was 70.2 ± 8.9 nm, indicating that the crystal grains were non-single-crystal.[26] The thicknesses of a-Sb_2Se_3 and c-Sb_2Se_3 films determined from cross-sectional SEM were 635 nm and 473 nm, respectively. The molar ratios between Sb and Se in a-Sb_2Se_3 and c-Sb_2Se_3 samples were 1:1.54 and 1:1.52 determined by energy-dispersive X-ray (EDX) spectroscopy, respectively.[26]

temperature dependent transmission spectrum

To confirm the optical transition type, TDTS of c-Sb_2Se_3 thin films were measured by recording the light intensities before and after passing through the sample which was mounted inside the helium-flow cryostat (Microstat MO, Oxford). The result is shown in Fig. 1(b). We divided the TDTS spectra into two regimes by grey dashed line (regions I and II). In region II (>150 K), film transmittance gradually decreased down to zero within 140 nm; while in region I (<150 K), there was a sharp drop in film transmittance within 50 nm. Such an observation suggested that Sb_2Se_3 is an intrinsically indirect bandgap semiconductor: at temperature above 150 K, both direct and indirect transitions contribute to film absorption because phonons are abundantly available to ensure the conservation of momentum for the electron to be pumped from valence band maximum to conduction band minimum; at temperature below 150 K, indirect transition is largely prohibited due to phonon bottleneck (the average number of phonons sharply decrease in inverse proportion to T^3),[27] leaving behind the only contribution from direct bandgap. The intermixing of indirect with direct absorption transition at temperatures above 150 K is also indicative of small energy difference between direct bandgap ($E_{g,\,d}$) and indirect bandgap ($E_{g,\,i}$).

At the high values of the linear absorption coefficient (10^4–$10^5\,cm^{-1}$) where the condition $\alpha d \approx 1$ is fulfilled, α could be calculated from $\alpha = -ln(T)/d$ (where T is the transmission percentage, d is the thickness of the Sb_2Se_3 film).[20] According to the Tauc's method,[28] the optical bandgap of c-Sb_2Se_3 at different temperature could be extracted using the following formula:

$$(\alpha h\nu)^{\gamma} = A(h\nu - E_g), \tag{1}$$

where α is the absorption coefficient, $h\nu$ is the photon energy, A is a constant related to effective mass, γ is equal to 2 and 1/2 for direct and indirect bandgap, respectively. By fitting $(\alpha h\nu)^2$ versus $h\nu$ curves linearly the direct bandgap could be retrieved from the horizontal axis intercept for film transmittance measured between 4.5 K and100 K (Fig. 1(c)). Similarly, indirect bandgap could be obtained by the linear fitting $(\alpha h\nu)^{1/2}$ versus $h\nu$ curves for the 150–300 K data and read the intercept (data not shown here). It should be noted that in the higher temperature range (region II), only indirect transition contributes to the photon absorption at the absorption onset (~1100 nm), justifying our fitting. The optical bandgaps as a function of temperature are shown in Fig. 1(d) with blue and green solid points representing experimentally derived $E_{g,\,d}$ and $E_{g,\,i}$ at different temperatures, respectively.

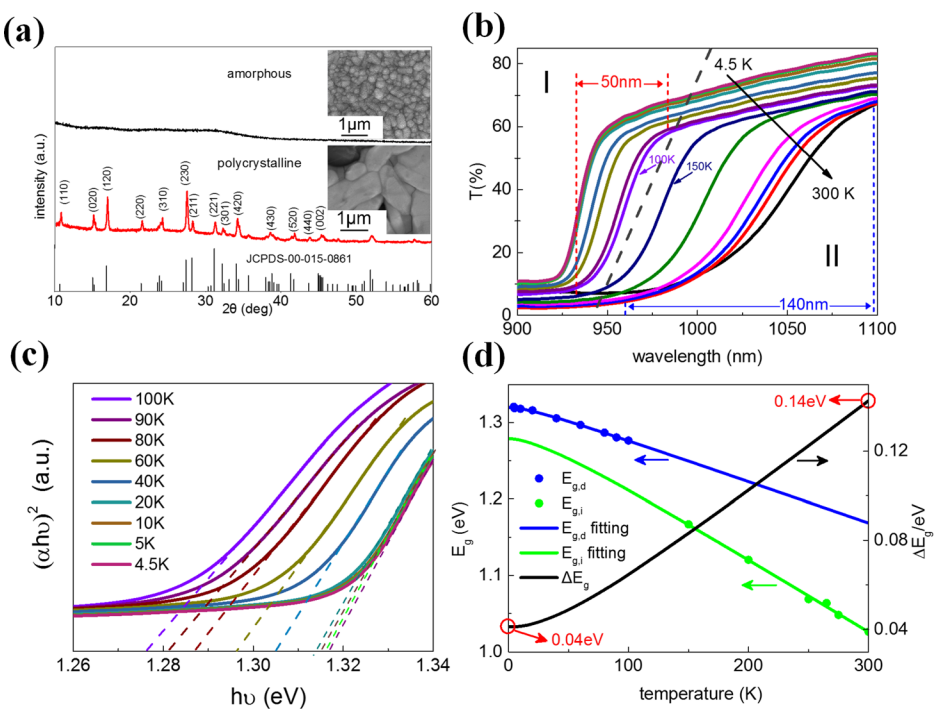

FIG. 1. (a) X-ray diffraction patterns of a-Sb_2Se_3 and c-Sb_2Se_3 thin films. Indexes of the diffraction peaks are also included for reference. Insets are the top-view SEM images of these Sb_2Se_3 films. (b) Temperature dependent transmission spectrum of c-Sb_2Se_3 thin films. The measured temperature (from low to high) were in the order of 4.5 K, 5 K, 10 K, 20 K, 40 K, 60 K, 80 K, 90 K, 100 K, 150 K, 200 K, 250 K, 265 K, 275 K, and 300 K. (c) Fitting of the absorption using direct bandgap assumption in the 4.5–100 K temperature range (region I). (d) Temperature dependent bandgap of c-Sb_2Se_3. Solid points are the experimental bandgaps at different temperatures, and the solid blue and green lines are the corresponding fitting curves for direct bandgap ($E_{g,\,d}$) and indirect bandgap ($E_{g,\,i}$), respectively. Dark line is the difference between $E_{g,\,d}$ and $E_{g,\,i}$.

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The temperature dependent E_g is fitted using the following empirical expression:[29]

$$E_g = E_g(0) - \alpha T^2/(T + \Theta_D), \quad (2)$$

where $E_g(0)$ is the energy bandgap at 0 K, α is a constant, and Θ_D is approximately the 0 K Debye temperature. The fitting results are shown as the solid lines in Fig. 1(d). Both α and Θ_D are used as fitting parameters and their values are shown in Table I. The parameters' uncertainties were given by fitting error.

Extending the temperature dependence of bandgaps curves from 0 K to 300 K utilizing the fitting formula we obtained the $E_{g,\ i}$(300 K) and $E_{g,\ d}$(300 K) as 1.03 ± 0.01 and 1.17 ± 0.02 eV, respectively. The differences between $E_{g,\ i}$ and $E_{g,\ d}$ at 0 K and 300 K are thus calculated to be 0.04 eV and 0.14 eV, respectively. Theoretical calculation also indicated that Sb_2Se_3 is indirect and the energy difference between $E_{g,\ d}$ and $E_{g,\ i}$ was approximately 0.02 eV at 0 K,[30] which was very close to our experimental value (0.04 eV). Due to the fairly small ΔE_g at 0 K, the valence electron lends priority to direct transition upon photon absorption, giving rise to the observed steep transmittance drop in region I (Fig. 2(b)) in our c-Sb_2Se_3 film. The intrinsically indirect bandgap of c-Sb_2Se_3 was further confirmed by TDPL measurement excited by a 960 nm laser in the 84–180 K temperature range (Fig. S2). Since the c-Sb_2Se_3 film is of high quality (when processed into CdS/Sb_2Se_3 solar cells, it could produce devices with short-circuit current densities (J_{sc}) above 25 mA/cm^2),[12] we thus believe the observed absence of photoluminescence even down to liquid N_2 temperature with strong excitation power is a further strong support to our TDTS observation that Sb_2Se_3 is an indirect bandgap semiconductor.

Like SnS, c-Sb_2Se_3 is indirect possibly due to the low symmetry crystal structures caused by the lone pair s electrons.[31] This is detrimental for photovoltaic application for poor absorption of photons with energy right above the indirect bandgap. On the other hand, the direct optical transition in c-Sb_2Se_3 is consisted of p->p transition (Se 4p orbitals to Sb 5p orbitals)[24,31] which enables very high absorption coefficient for photons with energy ~0.14 eV larger than the $E_{g,\ i}$. Optimal $E_{g,\ i}$ and $E_{g,\ d}$ values coupled with the strong absorption coefficient in direct transition make the light absorption not a serious problem in our Sb_2Se_3 thin film solar cells, as evidenced by the 25 mA/cm^2 J_{sc} in Sb_2Se_3 solar cells with only 390 nm absorber layer.[12] Nonetheless, the influences of indirect bandgap on the kinetics of photogenerated carriers and on other parameters of photovoltaic devices need further careful study.

We then tested the complex refractive index ($N = n + ik$, where N is the complex refractive index; n is the refractive index; and k is the extinction coefficient) of a-Sb_2Se_3 and c-Sb_2Se_3 film by ellipsometry (J.A.WOOLLAM, RC2) which

TABLE I. The fitting values of α and Θ_D in formula (2).

	$E_g(0)$ (eV)	α (10^{-4} K^{-1})	Θ_D(K)	E_g(300 K)
Direct	1.32 ± 0.01	6.06 ± 0.30	32.08 ± 0.34	1.17 ± 0.02
Indirect	1.28 ± 0.01	9.70 ± 0.41	45.00 ± 0.58	1.03 ± 0.01

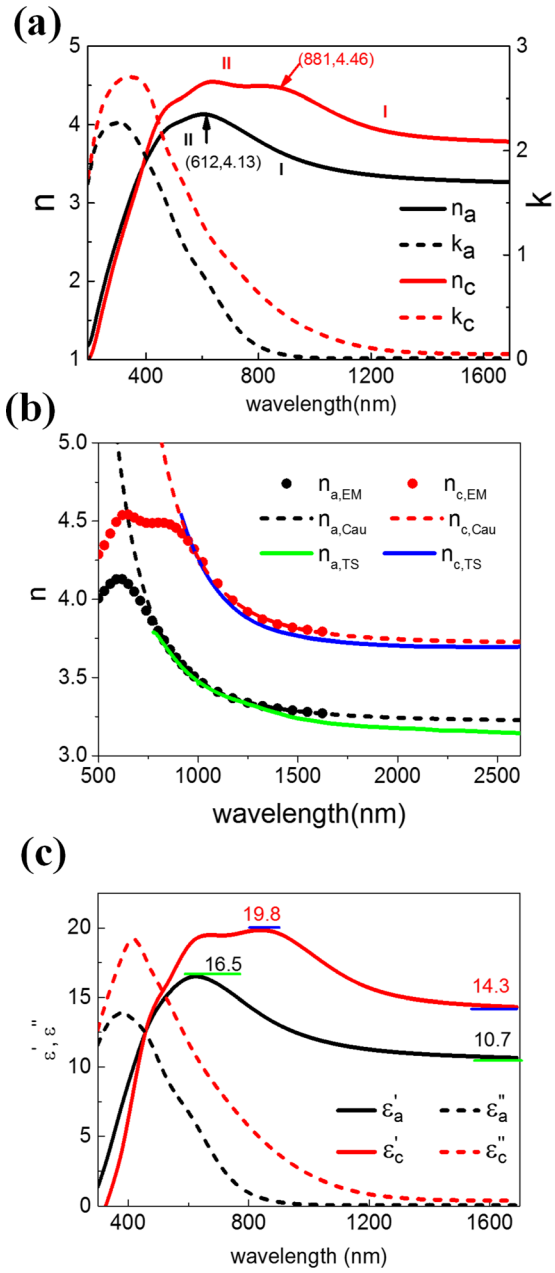

FIG. 2. (a) The refractive index and extinction coefficient of a-Sb_2Se_3 and c-Sb_2Se_3 films measured by ellipsometry in the wavelength range of 193–1693 nm. (b) The refractive index measured by ellipsometry (solid dots) and transmittance (solid lines) and fitted by Cauchy dispersion formula (dashed lines). (c) The complex dielectric constant of a-Sb_2Se_3 and c-Sb_2Se_3 films versus photon wavelength. The blue and green short solid lines mark the typical values of the real part of the relative dielectric constant.

is a precise and nondestructive instrument to record optical parameters through the analysis of phase difference and amplitude ratio between the incident and reflected polarized light.[32] The experimental data were analyzed by Complete EASE software. The measured refractive index and extinction coefficient of a-Sb_2Se_3 (n_a, k_a) and c-Sb_2Se_3 (n_c, k_c) are shown in Fig. 2(a). The apparent observation is that n is very large, approximately 4, for both a-Sb_2Se_3 and c-Sb_2Se_3 films in the wavelength of 500 nm and above. While similar refractive indices were observed for a-Sb_2Se_3 and c-Sb_2Se_3 films between 193 and 400 nm, c-Sb_2Se_3 shows higher refractive index than a-Sb_2Se_3 between 400 and 1693 nm. For

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extinction coefficient k, c-Sb_2Se_3 shows consistently stronger absorption than that of a-Sb_2Se_3 in the whole wavelength range possibly because annealing process decreased the structural imperfections and increased grain size.[33] The extinction coefficient is related to the absorption coefficient by $k = (\alpha\lambda)/(4\pi)$ (where α is the absorption coefficient, λ is the wavelength). Using the formula (1) with $\gamma = 1/2$, the indirect bandgaps of c-Sb_2Se_3 were calculated as 1.02 ± 0.01 eV (very close to $E_{g,\ i}(300\,\mathrm{K}) = 1.03 \pm 0.01$ eV from TDTS) and the optical edge of a-Sb_2Se_3 was calculated as 1.39 ± 0.02 eV.

The refractive index n_a and n_c could be divided into two parts (I and II) by peak points at 612 nm and 881 nm, respectively. In both part I, the non-absorbing region ($k \approx 0$), refractive index n_a and n_c could be fitted using the normal Cauchy's dispersion formula ($n = n_0 + a/\lambda^2 + b/\lambda^4$, where n_0, a, and b are constants and λ is the wavelength in nm).[34] The resulted equations are as follows:

$$\begin{cases} n_a \approx 3.21 + 7.51 \times 10^4/\lambda^2 + 1.98 \times 10^{11}/\lambda^4 \\ n_c \approx 3.69 + 3.3 \times 10^4/\lambda^2 + 5.33 \times 10^{11}/\lambda^4. \end{cases} \quad (3)$$

The black and red dashed lines in Fig. 2(b) are the fitting curves utilizing the normal Cauchy's dispersion formula (3), showing a very good fitting to experimental results. Obvious deviation appears in part II due to the strong absorption which undermines the application of Cauchy's fitting. We have further verified our ellipsometry results by Swanepoel's envelope method (Fig. S3).[21] Transmittance spectra of a-Sb_2Se_3 and c-Sb_2Se_3 films were recorded in the wavelength range of 400–2615 nm using UV/VIS spectrometer (Lambda 950, Perkin Elmer). As-obtained refractive index values of a-Sb_2Se_3 and c-Sb_2Se_3 ($n_{a,}$ and n_c) are shown in Fig. 2(b) with green and blue solid curves, respectively. The general trends and exact values of the refractive index obtained from transmittance spectrum agreed well with the values from ellipsometry measurement, further proving the credibility of our results.

The dielectric constant related to dielectric polarization and exciton binding energy is an essential parameter in semiconductors. The complex relative dielectric constant could be described by the follow formula:[34]

$$\begin{aligned} \varepsilon' &= n^2 - k^2, \\ \varepsilon'' &= 2nk, \end{aligned} \quad (4)$$

where ε' and ε'' are the real and imaginary parts of the complex relative dielectric constant; n and k are the refraction index and the extinction coefficient, respectively. The calculated complex dielectric constant is showed in Fig. 2(c). ε' ranged from 10.7 to 16.5 at 612–1693 nm for a-Sb_2Se_3 and 14.3 to 19.8 at 852–1693 nm for c-Sb_2Se_3, as marked with green and blue lines in Fig. 2(c). Compared with copper indium gallium diselenide (CIGS),[35] the larger dielectric constant of c-Sb_2Se_3 suggests smaller exciton binding energy and possibly immediate separation of electrons and holes upon photogeneration. The large ε'' related to the dielectric loss is advantageous for light harvesting in Sb_2Se_3 films.[36]

For photovoltaic devices, minimum absorber layer thickness for full incident light absorptipon is decided by the absorption coefficient. We present the penetration depth of a few photons with typical wavelength in Fig. 3(a). For photons below 800 nm wavelength, they are completely absorbed within 400 nm thick film; while for photons with wavelength beyond 900 nm, their penetration depth reaches to a micron. Indirect optical transition in c-Sb_2Se_3 accounts for the long penetration depth for photons at 900–1200 nm. Fortunately, the light intensity of the solar spectrum mostly distributes in the visible where the absorption coefficient is also high for c-Sb_2Se_3 thin films, so we can achieve a high photocurrent.

Besides, incident light reflectivity is also highly relevant for photovoltaic application and could be calculated using the following relation:[34]

$$R = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2}, \quad (5)$$

where n and k are the refraction index and extinction coefficient. Fig. 3(b) shows the calculated normal-incidence reflectivity from air to a-Sb_2Se_3, air to c-Sb_2Se_3, and CdS to c-Sb_2Se_3 (the refractive index of CdS comes from internet[37]). Direct illumination onto both c-Sb_2Se_3 and a-Sb_2Se_3 suffers roughly 40% reflection loss; while CdS is deposited on top, the reflection loss dropped to approximately 10% in the wavelength range of 500–1700 nm, thereby necessitating the application of anti-reflection layer to reduce reflectance induced photocurrent loss. We further solidified our conclusion by the experimental measurement of reflectance at air/

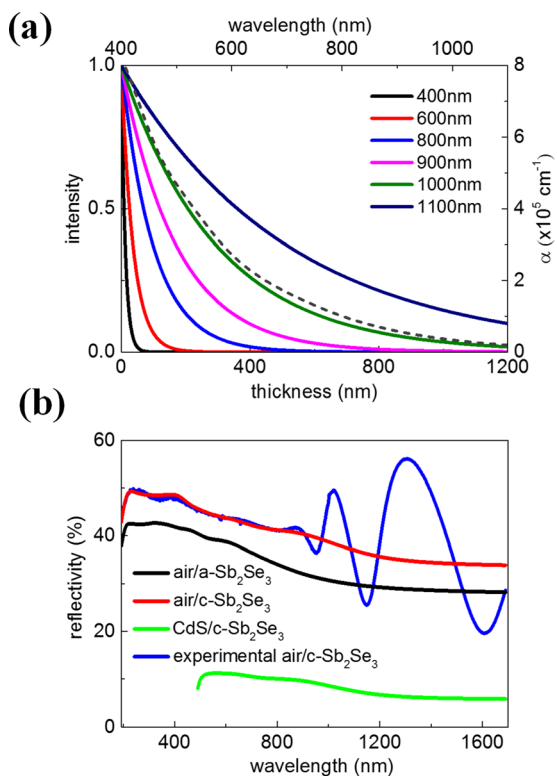

FIG. 3. (a) The incident light intensity decay versus wavelength. The dash grey line is the absorption coefficient (top x-axis; right y-axis). (b) The calculated normal-incidence reflectivity from air to a-Sb_2Se_3 (black line) and c-Sb_2Se_3 film (red line) and from CdS to c-Sb_2Se_3 film (green line). The solid blue line is the experimentally measured reflectance at the air/c-Sb_2Se_3 interface.

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c-Sb_2Se_3 interface by ellipsometry characterization, as evidenced by the perfect overlapping between calculated and measured results between 193 and 900 nm wavelength range [blue line, Fig. 3(b)].

In summary, we found that our c-Sb_2Se_3 film is intrinsically indirect bandgap material with the lowest $E_{g,\ i}$ of 1.03 ± 0.01 eV and the $E_{g,\ d}$ of 1.17 ± 0.02 eV at 300 K, which is supported by temperature dependent transmission spectrum, temperature dependent photoluminescence, and theoretical simulation. The optical edge of a-Sb_2Se_3 was measured as 1.39 ± 0.02 eV by ellipsometry. The basic optical constants of a-Sb_2Se_3 and c-Sb_2Se_3 such as complex refractive index, complex relative dielectric constants, and reflectivity were obtained in the 193–2615 nm wavelength range through a combined ellipsometry and transmittance measurements. The measured large refractive and dielectric values suggest that exciton binding energy in c-Sb_2Se_3 should be relatively small and an antireflective coating is recommended to enhance the light absorption of Sb_2Se_3 thin film solar cells.

This work was supported by the director fund of Wuhan National Laboratory for Optoelectronics, "National 1000 Young Talents" project and National Natural Science Foundation of China (NSFC 61322401, 91433105). Professor Shiyou Chen at East China Normal University is acknowledged for helpful discussions. The authors would like to thank the Analytical and Testing Center of HUST and the Center for Nanoscale Characterization and Devices of WNLO for the characterization support.

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